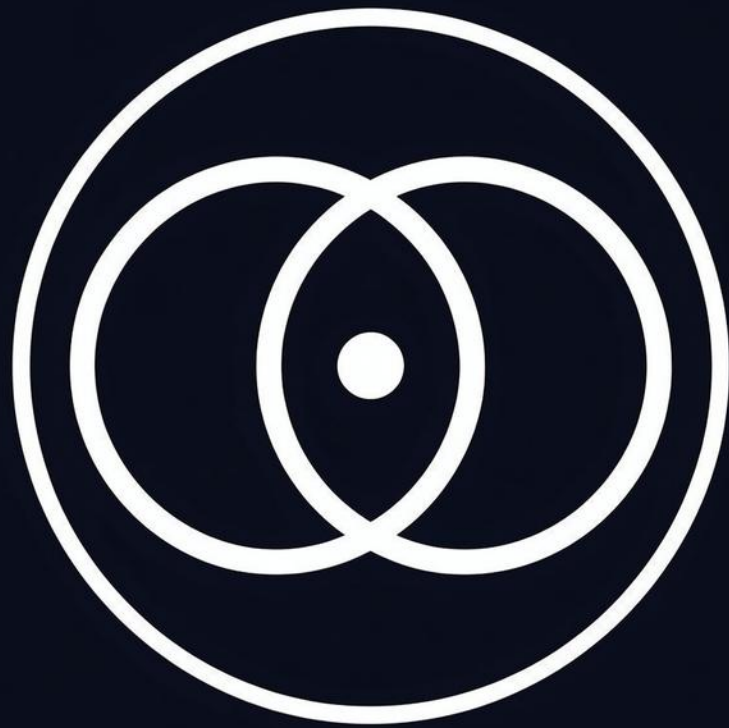




P29

Test-Time Scaling of Multi-Agent Collaboration: The Diversity Axis and the Decomposition Axis, with TUMIX and MAKER

测试时规模化协同：工具集成与极端分解投票

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Test-Time Scaling of Multi-Agent Collaboration: The Diversity Axis and the Decomposition Axis, with TUMIX and MAKER

Working Title (EN): Test-Time Scaling of Multi-Agent Collaboration: The Diversity Axis and the Decomposition Axis, with TUMIX and MAKER

Mid-volume engineering paper P29 • splitting "test-time scaling" into two orthogonal axes, **diversity** and **decomposition** • connecting to P9 quality saturation law • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: why does not training the model, only running it a few more times at inference, make it stronger? Three 2025–2026 works split the answer into two things—TUMIX (arXiv:2510.01279) proves **diversity** (parallel use of different tool strategies) effective; MAKER (arXiv:2511.09030) proves **extreme**

decomposition + stepwise voting (cutting million-step tasks to microagents, majority voting at each step to correct) can achieve zero errors. This paper argues these are two **orthogonal** scaling axes, not two ways of saying the same thing; and UDOS P9's "independent information" condition is the hidden premise common to both axes.

Evidence caliber (stated once for the whole paper): this is a method synthesis paper, **containing no new UDOS measured numbers**. Three external papers' existence verified online 2026-09-21; TUMIX's "+3.55%," MAKER's "about 1,048,575 steps zero errors," Thinking vs Doing's "Gemma 3 12B reaches open-source SOTA on WebVoyager/WebArena" all **paper reports**, not independently rechecked; this paper uses them to show "which axis is acting," not taking numbers as UDOS expected gains. UDOS P9 quality saturation law traced to v7.5.0.

Abstract

Test-time scaling is the third axis of 2025–2026 large-model capability gains—not changing weights, not piling parameters, only spending more compute at inference. This paper argues: multi-Agent test-time scaling actually has two **orthogonal** axes. The first is the **diversity axis**: running multiple Agents in parallel, each adopting different tools/strategies, then ensembling. TUMIX (arXiv:2510.01279) works along this axis—parallel multi-Agents using different tool strategies, iteratively sharing refinement, on Gemini-2.5-Pro average accuracy up to +3.55%, at near-equal inference cost (paper report). The second is the **decomposition axis**: extremely decomposing one long task into focused microagents, using majority voting at each step to correct. MAKER (arXiv:2511.09030) works along this axis—extreme decomposition + k=3 majority voting, first completing an about 1,048,575-step (million-scale) LLM task with zero errors (paper report). Thinking vs Doing (arXiv:2506.07976, NeurIPS 2025) suggests a third, different axis—the **interaction length axis**: through curriculum-style online RL adaptively adjusting the number of Agent-environment interaction rounds, using only a 12B Gemma 3 to reach open-source SOTA on WebVoyager/WebArena (paper report). This paper's core argument: the diversity axis is constrained by P9's quality saturation law—multiple parallel Agents if tool strategies isomorphic, carrying no independent information, ensemble gains quickly saturate; the decomposition axis is constrained by "whether voting can correct"—k=3 majority voting only effective when single-step error rates independent and below threshold, under common-cause faults voting fails. The two axes hence not "more Agents better," but a joint design of "diversity × decomposition depth." This paper gives UDOS's engineering specification for the two axes, effects pending [RESULT NEEDED].

Keywords: test-time scaling; multi-Agent ensemble; tool mixture; extreme decomposition; majority voting; independent information; P9; TUMIX; MAKER

Structured Abstract (background problem → method/argument → evidence → contribution)

- **Background problem:** why does test-time scaling strengthen? Industry often treats "running a few more Agents" as a single lever, but this hides two orthogonal mechanisms.
- **Core argument:** multi-Agent test-time scaling = diversity axis (heterogeneous tool strategy ensemble) × decomposition axis (extreme decomposition + stepwise voting correction), plus interaction length axis; P9's "independent information" is the hidden premise common to the first two axes.

- **Evidence lines:** (1) TUMIX tool mixture +3.55% (arXiv:2510.01279, paper report); (2) MAKER million-step zero-error extreme decomposition (arXiv:2511.09030, paper report); (3) Thinking vs Doing interaction length scaling (arXiv:2506.07976, paper report); (4) P9 quality saturation law $mse(k)=a+b/k$ (cpu-proto).
- **Contribution:** splitting test-time multi-Agent scaling into orthogonal dual axes; pointing out each axis's failure conditions; connecting the two axes to P9's independent information requirement.

1 Problem: Test-Time Scaling Is Not a Single Lever

1.1 The Third Scaling Axis

Beyond training-time scaling (piling parameters, data, compute), 2025–2026 saw a third axis: **test-time scaling**—freezing weights, only spending more compute at inference. This axis especially key for multi-Agent systems, because it directly asks "why is running multiple Agents in parallel useful."

1.2 Two "Multi-Agents" Often Conflated

Intuitively "running a few more Agents to vote" and "cutting the task into many small steps" seem like two ways of saying the same thing. This paper argues they are not:

- **Diversity axis:** same task, parallel multiple solutions, then ensemble. Variable "whether parallel Agents' strategies are heterogeneous."
- **Decomposition axis:** single task, serial step cutting, stepwise correction. Variable "how finely the task is cut, how each step corrects."

One spreads horizontally, one cuts vertically. They are orthogonal, can stack, and each has failure conditions.

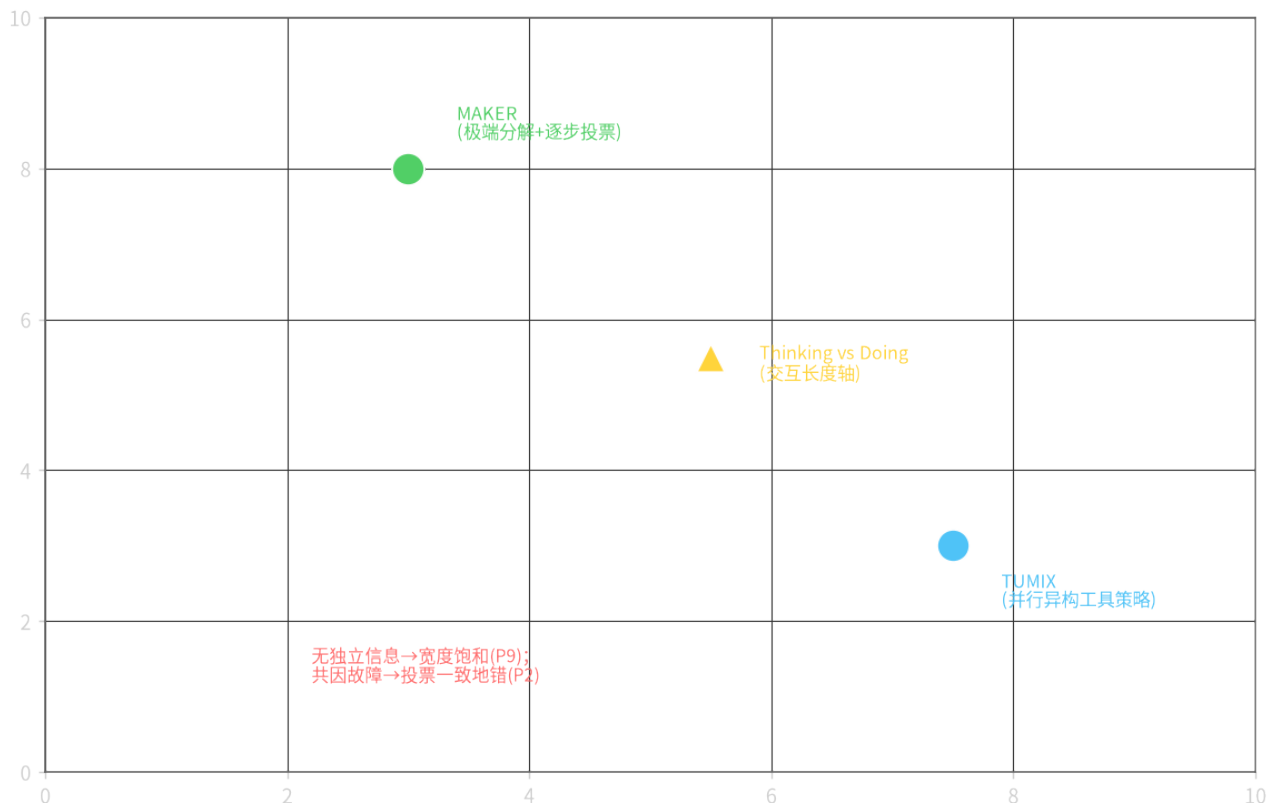


Figure 1 Horizontal diversity (parallel heterogeneous Agents), vertical decomposition depth (task cut granularity + stepwise voting); three representative works each occupy positions. Conceptual schematic, not measured data.

2 The Diversity Axis: TUMIX and the Independent Information Condition

2.1 What TUMIX Did

TUMIX (arXiv:2510.01279, TUMIX: Multi-Agent Test-Time Scaling with Tool-Use Mixture) runs multiple Agents in parallel, each adopting **different tool strategies**, then iteratively sharing refinement. The paper reports on Gemini-2.5-Pro/Flash average accuracy over the best baseline up to **+3.55%**, at near-equal inference cost (paper report, not independently rechecked).

Its key is not "running more," but "**each using different tools**." If parallel Agents use exactly the same tools and prompts, they give highly correlated answers—precisely the homogeneous copies P9 warns about.

2.2 P9 Condition: Diversity Must Carry Independent Information

P9's quality saturation law ($mse(k)=a+b/k$, cpu-pto) writes the diversity axis's cost fixed. For an ensemble of k members' mean squared error:

$$mse(k)=a+\frac{b}{k}$$

where a is irreducible error, b set by inter-member **independence**. TUMIX's +3.55% holds precisely because "different tool strategies" raised b (member independence); if tool strategies made homogeneous, b collapses, adding more parallel Agents only votes the same wrong answer k times.

Plain language: **the diversity axis's gain ceiling depends on how different your parallel Agents actually are, not how many there are.**

3 The Decomposition Axis: MAKER and Stepwise Voting Correction

3.1 What MAKER Did

MAKER (arXiv:2511.09030, Solving a Million-Step LLM Task with Zero Errors) **extremely decomposes** (MAD, Massive Decomposition) one super-long task into focused microagents, each step using multi-Agent **majority voting** to correct, and discarding red-flagged correlated errors. The paper reports this is the first system completing an over-million-LLM-step task with **zero errors** (about 1,048,575 steps, $k=3$ majority voting; paper report, not independently rechecked).

Its key is not "cutting fine," but "**each step having independent correction**"—a single-point LLM's persistent error rate accumulates and explodes with steps; stepwise majority voting presses long-range error rates back to controllable.

3.2 When Voting Works, When It Fails

Majority voting's effectiveness has a classic premise: each step's errors **approximately independent**. If single-step error rate p , $k=3$ majority voting drops error rate to about $3p^2(1-p)+p^3$, significantly declining when $p<0.5$. But—

- **Under common-cause faults voting fails:** if all microagents are carried away by the same misleading input (the semantic common-cause fault P2 identifies), $k=3$ consistently errs, majority voting becoming "consistently erring more confidently." This is the decomposition axis's hard boundary.
- **Decomposition introduces cut errors:** cut too coarse, microagents still face long-range dependencies; cut too fine, inter-microagent handoff errors accumulate. Optimal decomposition granularity is a design parameter.

Plain language: **the decomposition axis turns "single-point error rate" into "stepwise error rate," but it cannot save the case where all Agents are deceived together.**

4 The Third Axis: Interaction Length (Thinking vs Doing)

Thinking vs. Doing (arXiv:2506.07976, NeurIPS 2025 SEA Workshop) works along the third axis—**test-time interaction length**: using curriculum-style online RL to adaptively adjust Agent-environment interaction rounds (exploration, backtracking, dynamic replanning), the paper reports only a 12B Gemma 3 reaching open-source/public-data SOTA on WebVoyager/WebArena (about +9%/+8% over non-fine-tuned Agents, paper report).

This axis is neither "parallel a few" (diversity) nor "cut a few steps" (decomposition), but "**how long the same Agent interacts back and forth with the environment.**" It reminds us: test-time scaling has at least three adjustable dimensions—width (parallel), depth (decomposition), duration (interaction).

Table 1 Three-axis comparison of test-time multi-Agent scaling

Axis	Variable	Representative work	Failure condition	Constrained by
diversity (width)	parallel Agent strategy heterogeneity	TUMIX (+3.55%)	tool strategies homogeneous → independence b collapses	P9 quality saturation law
decomposition (depth)	task cut granularity + stepwise voting	MAKER (million-step zero errors)	common-cause fault → voting consistently errs	P2 semantic common-cause fault
interaction (duration)	environment interaction rounds	Thinking vs Doing (Gemma3 12B)	interaction length non-adaptive → waste or underexploration	curriculum scheduling design

5 Dross / Failure Boundaries / Falsifiable Conditions

5.1 Dross (three misreadings deifying test-time scaling)

1. **+3.55% is not a universal gain.** TUMIX's gain depends on task type and baseline choice, switching tasks may be smaller or even disappear; do not take single-benchmark increments as "multi-Agent universal."

2. **Million-step zero errors $\neq$ long-task universal.** MAKER's zero errors built on highly structured tasks "each step independently judgeable right/wrong by microagents"; semantic acceptance work orders (with hidden ground truth) lack such stepwise oracles, stepwise voting unable to land.
3. **Cheap small-model SOTA $\neq$ small models strong enough.** Thinking vs Doing used 12B to reach open-source SOTA, relying on test-time interaction scheduling, not the model itself becoming smaller; reading this as "no need for large models" is a misreading.

5.2 Failure boundaries

- The diversity axis and P9 homogeneous copies are the same coin: **no independent information, width scaling ineffective.**
- The decomposition axis on tasks **without stepwise oracles** cannot correct—UDOS's semantic acceptance work orders precisely lack stepwise oracles, this is the biggest obstacle to the decomposition axis landing on UDOS.

5.3 Falsifiable conditions

- **F29-a:** if on UDOS work orders, changing parallel Agents' tool strategies from "heterogeneous" to "homogeneous," ensemble quality decline matches P9's $\$b\$$ collapse prediction, then this paper's "diversity = independent information" argument holds; if unrelated, needs correction.
- **F29-b:** if after introducing common-cause fault injection, the decomposition axis's majority voting correction advantage disappears, then the "voting pierced by common-cause faults" boundary holds.
- The above are post-hoc scorable criteria under the design proposal, UDOS actual numbers pending [RESULT NEEDED].

6 Conclusion

Test-time scaling is not the single knob "run a few more Agents." It has at least three axes: width (diversity), depth (decomposition), duration (interaction). TUMIX proves the width axis buys gains by "tool heterogeneity," MAKER proves the depth axis presses errors by "extreme decomposition + stepwise voting," Thinking vs Doing proves the duration axis raises performance by "adaptive interaction rounds." UDOS's contribution is giving each of these three axes failure conditions: width constrained by P9 independent information, depth constrained by P2 common-cause faults.

Test-time scaling is not "piling more Agents," but in the three directions diversity $\times$ decomposition $\times$ interaction, each confirming "these Agents truly carry independent information, truly can be corrected step by step." Piling people does not solve problems, piling with discrimination does.

References

Ensemble and test-time compute

1. Hansen, J. N., et al. (2024). Let's Verify Step by Step. OpenAI. (classic stepwise reasoning/process reward work)
2. Brown, T. B., et al. (2020). Language Models are Few-Shot Learners (test-time prompting background).

2026 frontier materials (existence verified online 2026-09-21; numbers paper reports, not independently rechecked)

1. TUMIX (2025). TUMIX: Multi-Agent Test-Time Scaling with Tool-Use Mixture. arXiv:2510.01279. (heterogeneous tool strategies parallel, Gemini-2.5-Pro up to +3.55%)
2. MAKER (2025). Solving a Million-Step LLM Task with Zero Errors. arXiv:2511.09030. (extreme decomposition + $k=3$ majority voting, about 1,048,575 steps zero errors)
3. Thinking vs. Doing (2025). Agents that Reason by Scaling Test-Time Interaction. arXiv:2506.07976, NeurIPS 2025 SEA Workshop. (Gemma 3 12B reaches open-source SOTA on WebVoyager/WebArena)

UDOS paper volume (this series, evidence grades marked in text)

1. UDOS v7.5.0 paper volume: P2 "BFT-lite Stop Decision" (semantic common-cause fault identification), P9 "Quality Saturation Law and Homogeneous Headcount Expansion Failure Boundary" ($mse(k)=a+b/k$, cpu-proto).
2. UDOS v7.6.0 paper volume: P20 "Heterogeneity Dividend and Homogeneous Saturation," P33 "v7.6.0 Architecture Master Outline."

Evidence Discipline and Reproduction Notes

- This paper is a **method synthesis paper**, no new experiments; three-axis division, failure conditions, F29-a/b criteria are analysis frameworks, actual gains on UDOS are [RESULT NEEDED], forbidden to write as measured.
- TUMIX +3.55%, MAKER million-step zero errors, Thinking vs Doing's Gemma3 12B SOTA all paper reports, not independently rechecked, only used to show "which axis is acting."
- $mse(k)=a+b/k$ traced to P9 (cpu-proto); common-cause fault boundary traced to P2.
- This paper does not claim test-time scaling can replace training scaling; three axes are parallel levers, not replacement relationships.